



Making This Versatile Arduino-Based Frequency Generator Is Simple

K. Padmanabhan and
A.K. Mariselvam

This is a simple function generator that works in the audio frequency range. It can be useful for amplifier testing, experimentation in digital signal processing (DSP) and electronics labs. The author's prototype is shown in Fig. 1.

Circuit diagram of the sine, square and ramp Arduino-based frequency generator is shown in Fig. 2. It is built around an Arduino Uno board (Board1), 16x2 LCD (LCD1), two 10-kilo-ohm potmeters (VR1 and VR2) and a few other components.

Potmeter VR1 connected to pin 3 of LCD1 is used for controlling the contrast of LCD1. VR2 connected to analogue input A0 pin of the Arduino Uno board is used to adjust time period of the output waveforms.

Three separate outputs are taken from pins 3, 9 and 10 of Arduino board. Pin 3 is for square wave, pin 9 for sine wave and pin 10 for ramp wave output. Pins 9 and 10 outputs are actually pulse width modulated (PWM) signals carrying the analogue signals, and are obtained after filtering with a simple resistor-capacitor filter circuit. The square wave output at pin 3 is directly taken without filter as

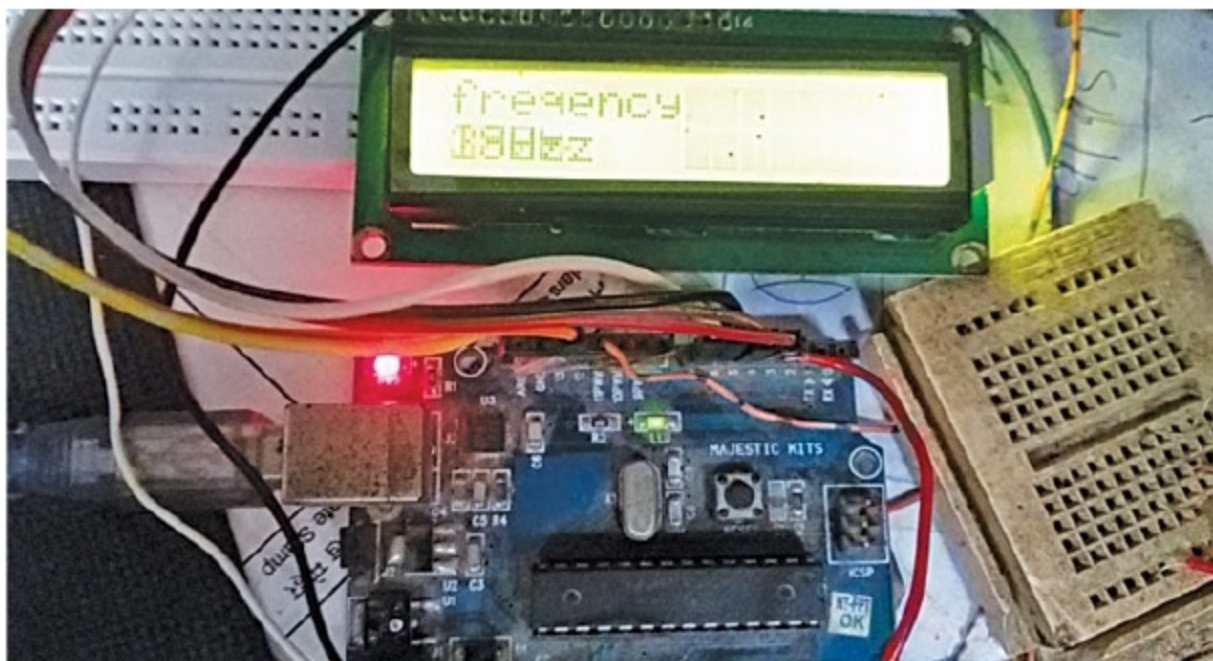
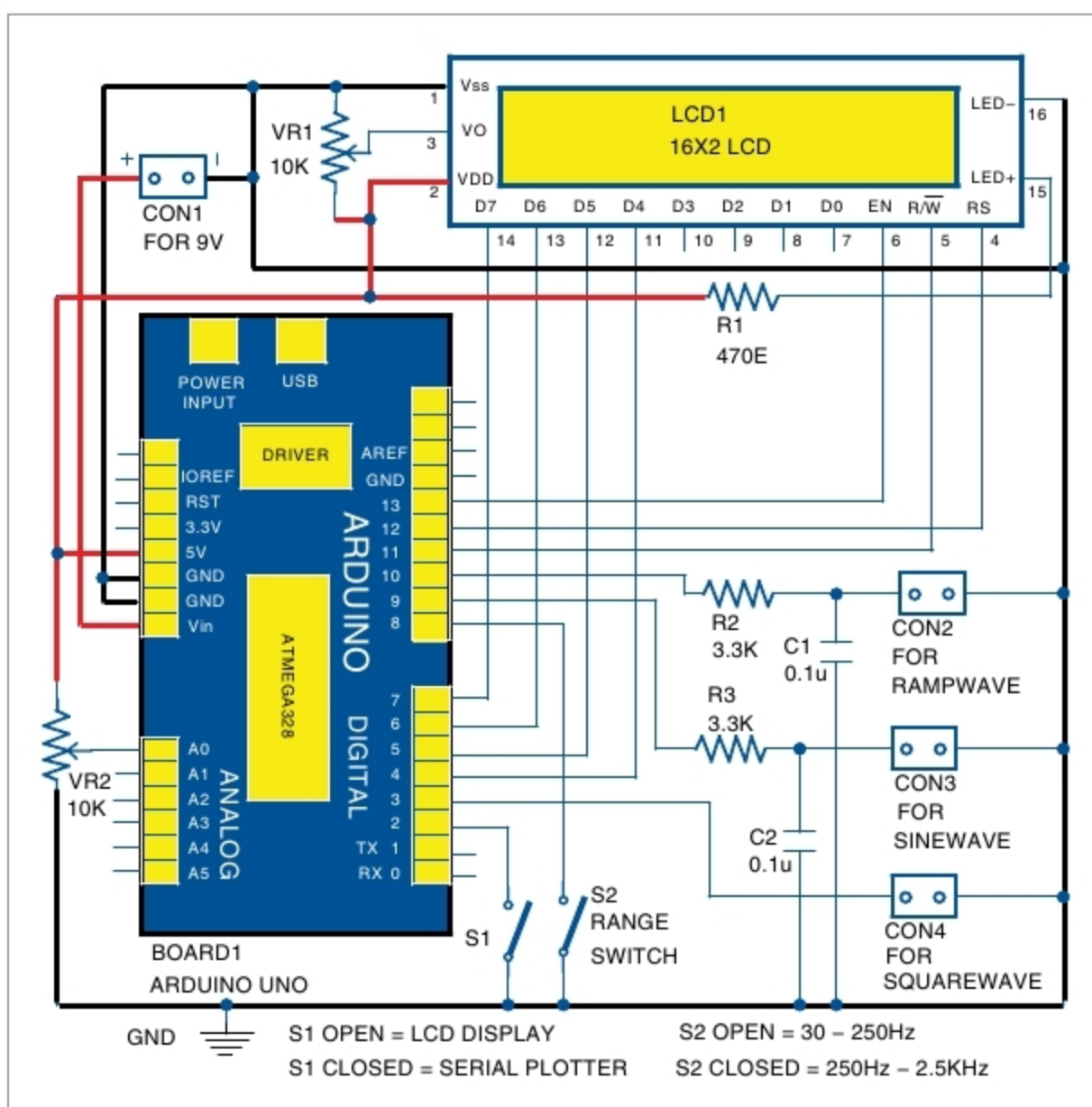


Fig. 1: Author's prototype



PARTS LIST

Semiconductors:

Board1 - Arduino Uno board

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1 - 470-ohm
R2, R3 - 3.3-kilo-ohm
VR1, VR2 - 10-kilo-ohm potmeter

Capacitors:

C1, C2 - 0.1 μ F ceramic disk

Miscellaneous:

CON1-CON4 - 2-pin connector
S1, S2 - Push to on switch
LCD1 - 16x2 LCD
- 9V battery/adaptor
- 17-pin Berg female connector for Board1
- 16-pin Berg female connector for LCD1
- Jumper wires

(All these are available from kitsnspares.com)

Fig. 2: Circuit diagram of Arduino based frequency generator